Evaluation Report — Federated Traffic-Flow Prediction with Communication-Cost Evaluation

Team NeuroForge · AIORI Hackathon 3 · PS-003 (A3-PS003-TC194)

All numbers below are produced by `scripts/report.py` from the logs in `results/runs/`. Test-period numbers come from the validation-selected checkpoint of each run, evaluated once.

Key findings

1. **FedAvg matches centralised accuracy with 30% of the bytes.** Test MAE: centralised 2.750, FedAvg IID 2.749, FedAvg non-IID 2.758 mph (3 seeds), for 6.00 MB of weights versus 19.86 MB of raw data.
2. **Geographic non-IID costs little; strong non-IID costs more.** Location-based clients add +0.009 mph over IID and need a few more rounds; clients split by speed level add +0.033 mph and never reach the 1% target in 50 rounds.
3. **INT8 update compression is almost free.** It cuts total traffic from 6.00 to 3.75 MB (37% less) for a +0.001 mph change in test MAE.
4. **Federation helps the intersections.** Training alone (zero bytes) gives 2.777 mph; joining FedAvg lowers it to 2.758 mph.
5. **FedAvg is not automatically cheaper.** A 50-round FP32 retraining costs as much as 25 days of raw streaming, so federated training saves bandwidth only if models are retrained less often than that.

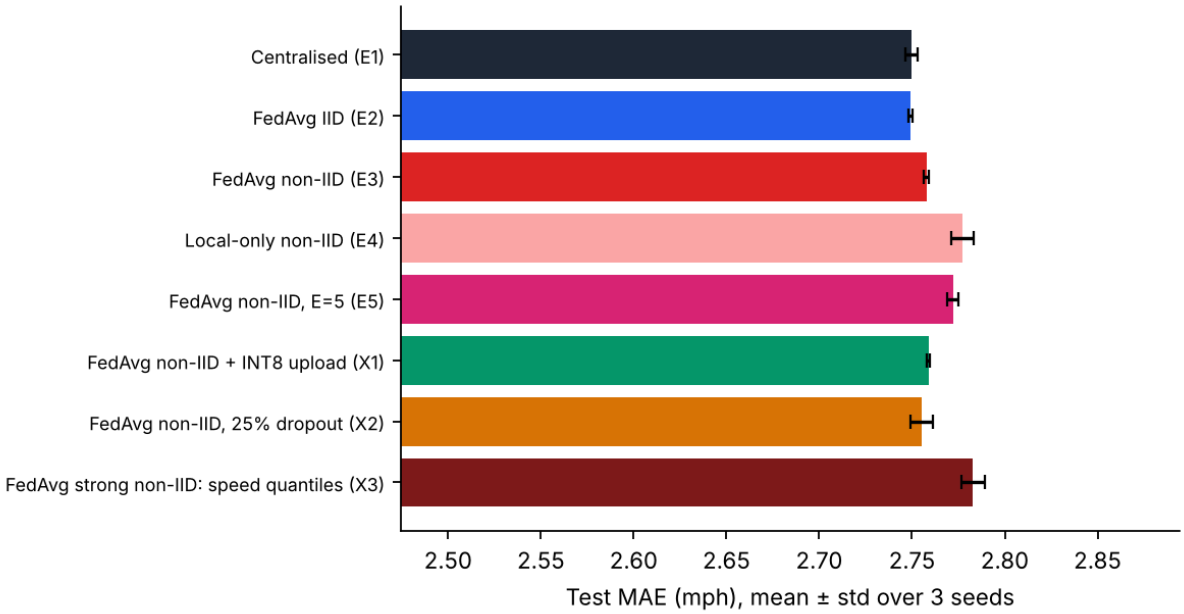
1. Setup

- Dataset: METR-LA, 34,272 five-minute steps × 207 loop detectors, 2012-03-01 → 2012-06-27; missing readings (0 mph) re-measured on load: **8.11%**. SHA-256 `64784b76d6fb8ec9…`.
- Chronological split 70/10/20: train 23,990 steps (to 2012-05-23 07:05:00), validation 3,427, test 6,855. z-score fitted on training readings only (μ = 58.58 mph, σ = 12.82 mph).
- Model: one GRU (hidden 32) + linear head, 3,747 parameters = 14,988 B in float32; input 12 steps × 4 channels; outputs +5/+10/+15 min; masked L1 loss.
- FedAvg: K = 4 clients, full synchronous participation, E = 1 local epoch, R = 50 rounds; centralised baseline trains R·E = 50 epochs on pooled windows (equal data passes).
- Learning rate chosen on validation from the grid {1e-3, 3e-3, 1e-2} (8-round trials, seed 11): centralised **0.01**, FedAvg **0.01** (Adam, gradient clip 1.0, batch 1,024). The selected value is the largest in the pre-declared grid; we did not widen the grid after seeing results.
- Seeds 11, 22, 33 control initialisation, IID assignment, k-means initialisation and batch order. Results are mean ± sample std (n = 3).
- Compute budget (declared deviation): training windows use one anchor every 24 steps (120 min), per-round validation every 6 steps; the **test period uses every window**. All methods use the same strides, so the comparison stays fair.

2. Main results (RQ1, RQ2, RQ5)

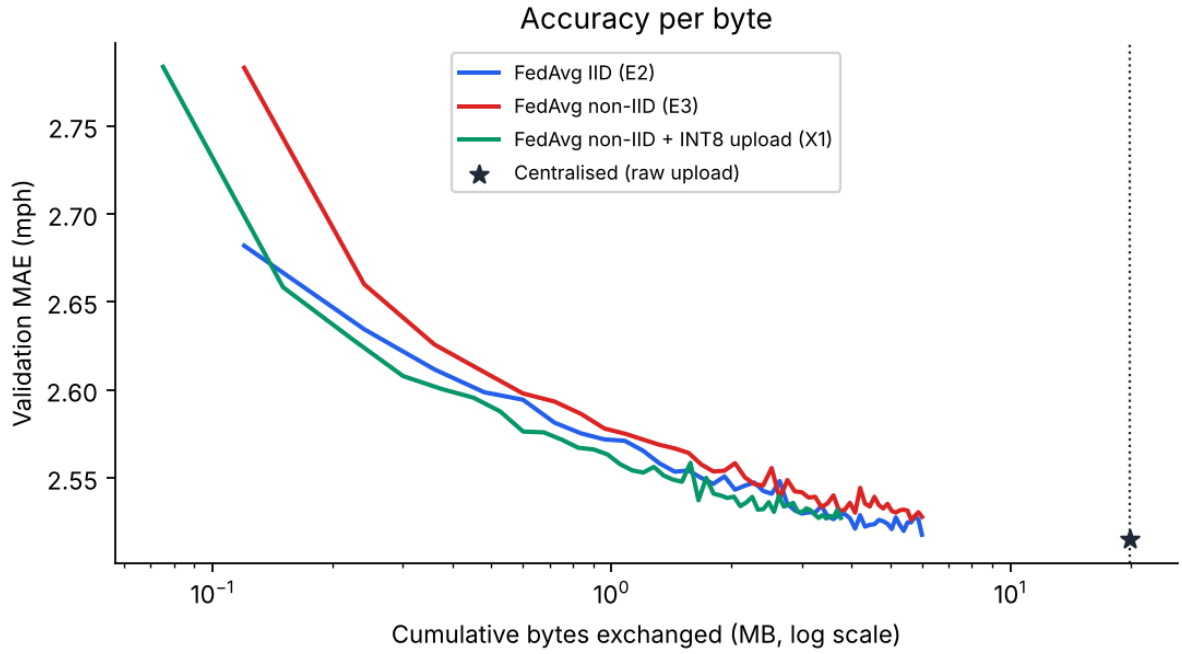
Method	Test MAE (mph)	Test RMSE (mph)	MAE @ +15 min	RMSE @ +15 min	Bytes exchanged
Centralised (E1)	2.750 ± 0.003	5.219 ± 0.017	3.049 ± 0.005	5.964 ± 0.021	19.86 MB
FedAvg IID (E2)	2.749 ± 0.001	5.226 ± 0.009	3.048 ± 0.003	5.980 ± 0.011	6.00 MB
FedAvg non-IID (E3)	2.758 ± 0.001	5.232 ± 0.007	3.061 ± 0.001	5.980 ± 0.008	6.00 MB
Local-only non-IID (E4)	2.777 ± 0.006	5.271 ± 0.009	3.077 ± 0.009	6.025 ± 0.013	0 (no communication)
FedAvg non-IID, E=5 (E5)	2.772 ± 0.003	5.247 ± 0.014	3.075 ± 0.007	5.991 ± 0.015	1.20 MB
FedAvg non-IID + INT8 upload (X1)	2.759 ± 0.001	5.238 ± 0.003	3.061 ± 0.000	5.989 ± 0.006	3.75 MB
FedAvg non-IID, 25% dropout (X2)	2.755 ± 0.006	5.221 ± 0.014	3.056 ± 0.007	5.971 ± 0.016	5.26 MB
FedAvg strong non-IID: speed quantiles (X3)	2.783 ± 0.006	5.259 ± 0.018	3.091 ± 0.012	6.000 ± 0.022	6.00 MB

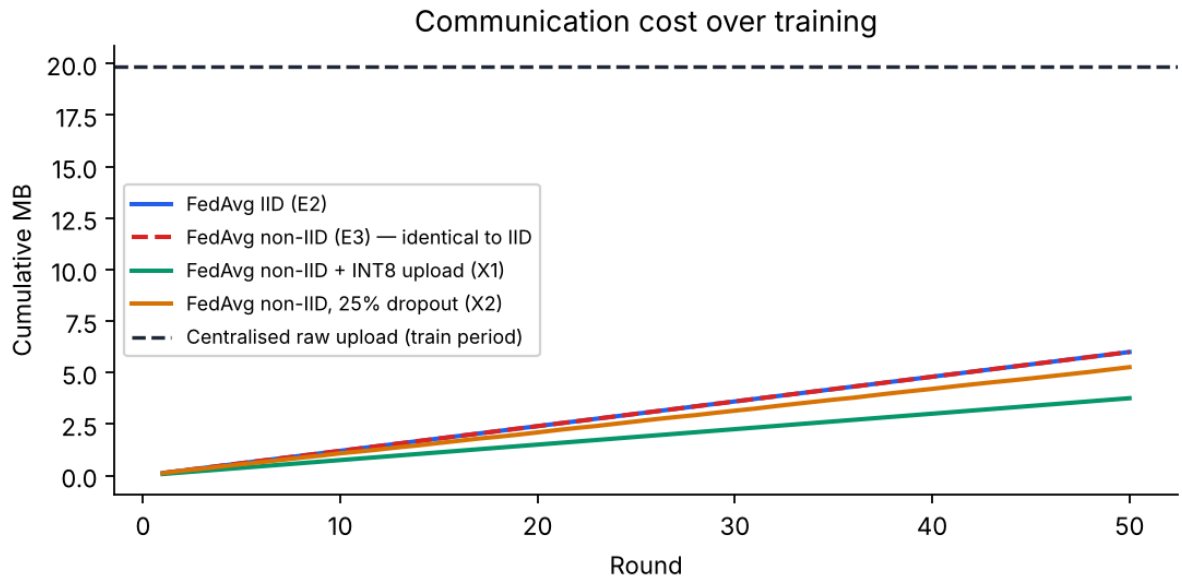
- **RQ1 — gap to centralised.** FedAvg-IID is ±0.000 mph MAE (±0.0%) and FedAvg-non-IID +0.008 mph (+0.3%) relative to centralised training. RMSE gaps: +0.007 and +0.013 mph. In words: FedAvg matches centralised accuracy to within a few hundredths of a mph while exchanging 30% of the bytes.
- **RQ2 — non-IID effect.** ΔMAE (non-IID − IID) = +0.009 mph, ΔRMSE = +0.006 mph. Rounds to reach τ = 1.05 × best centralised validation MAE: IID [2, 2, 3], non-IID [3, 3, 3] (per seed). With a stricter 1% target (τ_1 = 1.01 × best centralised, added by us because the 5% target is reached almost immediately): IID [21, 17, 20], non-IID [27, 18, 22] ('None' = not reached in 50 rounds). Validation-curve volatility (sum of upward steps): IID 0.096 ± 0.009, non-IID 0.110 ± 0.025.
- **RQ5 — consistency.** Method ordering by test MAE per seed: seed 11: Centralised < FedAvg IID < FedAvg non-IID; seed 22: FedAvg IID < Centralised < FedAvg non-IID; seed 33: FedAvg IID < Centralised < FedAvg non-IID. FedAvg non-IID has the highest error in every seed; centralised and FedAvg IID swap places, which means their difference is within seed-to-seed noise.

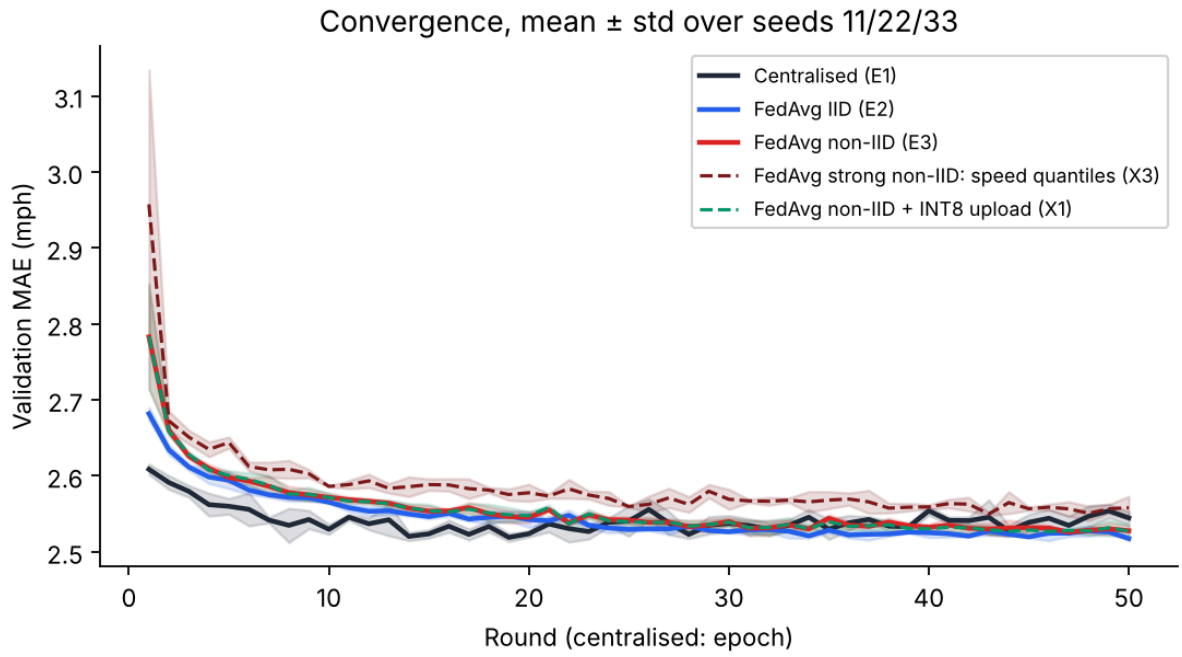


3. Communication cost (RQ3, RQ4)

- Analytic per-round payload (K = 4): 4 × 14,988 B down + 4 × 14,996 B up = **119,936 B**. The simulator's logged bytes equal this value exactly in every full-participation round (unit test `test_logged_bytes_equal_formula`). IID and non-IID runs send identical bytes per round (H3 confirmed), because payload depends only on model size.
- 50 rounds = 6.00 MB (+ 96 B one-off z-score statistics). Centralised training instead uploads every training reading once: 207 × 23,990 × 4 B = **19.86 MB**, i.e. the cost of ≈166 FedAvg rounds. In continuous operation, raw streaming costs 238.5 kB per day forever, whereas FedAvg pays only when it retrains.
- **Bytes-to-target** (first round whose validation MAE ≤ 1.05 × best centralised): IID 0.28 MB (reached in 3/3 seeds), non-IID 0.36 MB (reached in 3/3 seeds), non-IID + INT8 0.23 MB (reached in 3/3 seeds).
- **Bytes-to-target at 1%:** IID 2.32 MB (3/3 seeds), non-IID 2.68 MB (3/3 seeds), FedAvg non-IID + INT8 upload 1.70 MB (3/3 seeds), FedAvg strong non-IID: speed quantiles not reached. For comparison, centralised training needs the full 19.86 MB raw upload before its first epoch.







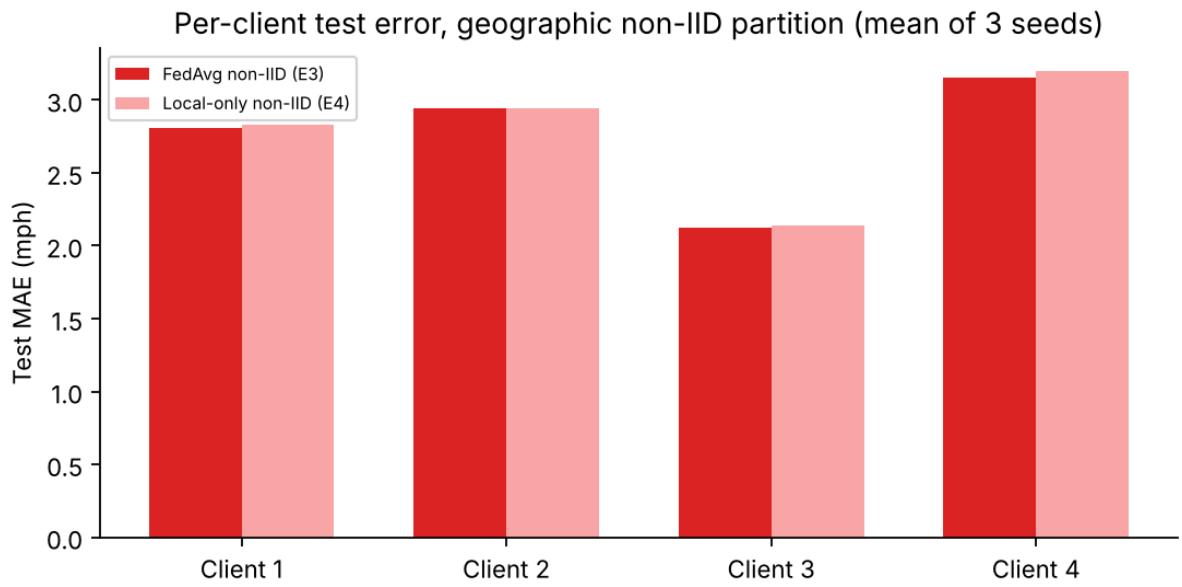
4. Extensions (beyond the mandated scope, reported separately)

- **FedAvg non-IID + INT8 upload (X1)** — test MAE 2.759 ± 0.001 mph (+0.001 vs FedAvg non-IID), bytes 3.75 MB, simulated link time 7.4 s.
- **FedAvg non-IID, 25% dropout (X2)** — test MAE 2.755 ± 0.006 mph (-0.003 vs FedAvg non-IID), bytes 5.26 MB, simulated link time 9.6 s.
- **FedAvg strong non-IID: speed quantiles (X3)** — test MAE 2.783 ± 0.006 mph (+0.025 vs FedAvg non-IID), bytes 6.00 MB, simulated link time 9.6 s.
- **FedAvg non-IID, E=5 (E5)** — test MAE 2.772 ± 0.003 mph (+0.014 vs FedAvg non-IID), bytes 1.20 MB, simulated link time 1.9 s.

- **Local-only non-IID (E4)** — test MAE 2.777 ± 0.006 mph (+0.019 vs FedAvg non-IID), bytes 0.00 MB, simulated link time 0.0 s.

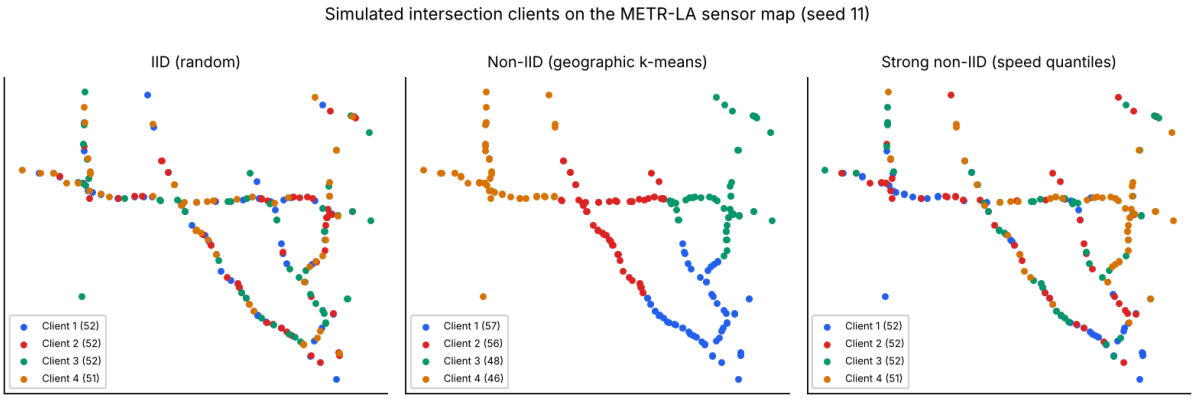
- X1 compresses only the *upload* (update delta w_k − w_t); downloads stay float32. X2 drops each client after download with probability 0.25 per round. X3 uses the proposal's documented fallback partition (quantiles of training mean speed) as a stress test. Link-time figures use a 2 Mbit/s uplink, 10 Mbit/s downlink, 60 ms RTT model and are illustrations, not measurements.

- **Who benefits from federation?** Per-client test MAE, FedAvg vs local-only (non-IID, 3-seed mean): client 1: 2.809 vs 2.828, client 2: 2.946 vs 2.945, client 3: 2.122 vs 2.142, client 4: 3.151 vs 3.195. FedAvg is better for client(s) [1, 3, 4]; a client whose own data is enough can lose slightly from sharing one global model, which is the classic personalisation trade-off.



5. Client heterogeneity check (seed 11, training period)

Partition	Client	Sensors	Mean speed	Std	Share < 40 mph
iid	1	52	58.97	12.77	9.6%
iid	2	52	58.32	12.8	9.9%
iid	3	52	58.42	12.95	11.4%
iid	4	51	58.62	12.76	11.0%
noniid	1	57	55.29	15.27	18.0%
noniid	2	56	60.05	11.88	7.9%
noniid	3	48	62.35	7.69	2.1%
noniid	4	46	57.04	13.51	12.9%
speed	1	52	50.28	15.76	24.6%
speed	2	52	57.66	12.11	10.0%
speed	3	52	61.52	9.3	4.8%
speed	4	51	65.12	7.23	2.2%



6. Byte-budget decision support (seed 11, non-IID)

Byte budget	Best FP32 checkpoint	Best INT8 checkpoint
0.5 MB	round 4, val MAE 2.610	round 6, val MAE 2.600
1.0 MB	round 8, val MAE 2.578	round 13, val MAE 2.557
2.0 MB	round 15, val MAE 2.550	round 24, val MAE 2.539
6.0 MB	round 50, val MAE 2.520	round 50, val MAE 2.521

7. Limitations (stated honestly)

- METR-LA holds aggregated 5-minute highway speeds, not trip logs, and its sensors are highway detectors, not signalised junctions; privacy is a simulated motivation, not a property of this data.
- Sharing model updates is not a privacy guarantee (gradient-leakage attacks exist); no differential privacy or secure aggregation is implemented.
- Byte counts are tensor payloads; protocol headers, TLS/TCP framing and retransmissions are excluded for every method.
- Training windows were subsampled for the CPU budget; absolute errors would likely fall slightly with every window, for all methods alike.
- n = 3 seeds: mean ± std are descriptive; no significance claims are made.

Reproducibility: Python 3.13.17, PyTorch 2.14.1+cu130, NumPy 2.5.3, scikit-learn 1.9.1. Re-create everything with `python scripts/run_experiments.py --suite all && python scripts/report.py`.